



INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

**NYKAA PRODUCT ANALYSIS USING WEB SCRAPING,
DATA CLEANING & EXPLORATORY DATA ANALYSIS (EDA)**

**Presented By -
B.LAHARI**



About me

- **Name: B. Lahari**
- **Education Background: Graduated B.Tech [CSE-AIML]**
- **Why I Want to Become a Data Scientist:**
 - I want to become a data scientist because I enjoy working with data, finding patterns, and solving real-world problems.
 - As a BTech student, I am interested in using my technical skills to turn data into meaningful insights and smart decisions.
- **Work Experience:**
 - I am a fresher, currently focusing on learning and building practical skills

Agenda (This should be the PPT flow)

- Introduction
- Business Understanding
- Problem Statement
- Objectives
- Tools & Technologies
- Methodology
- Data Cleaning
- Exploratory Data Analysis
- Power BI Dashboard
- Key Insights
- Conclusion
- Future Scope

Introduction

- Nykaa is one of India's leading beauty and personal care e-commerce platforms that offers thousands of products across multiple brands and categories.
- Due to the large volume of product information, manually analyzing prices, discounts, ratings, and reviews becomes difficult.
- This project automates data collection using web scraping and performs data analysis to discover useful business insights.

Business Understanding

- Businesses need to understand customer preferences, pricing strategies, discounts, and product popularity to improve sales and customer satisfaction.

Problem Statement

- Nykaa offers thousands of products across multiple categories. Manual analysis of pricing, discounts, and customer reviews is time-consuming. This project automates data collection and analysis using web scraping and visualization.
- An automated system is required to collect, clean, analyze, and visualize Nykaa product data for better business decision-making.

Objectives

- Scrape product information from Nykaa
- Create a structured dataset
- Clean the collected data
- Perform Exploratory Data Analysis
- Identify pricing and rating trends
- Develop an interactive Power BI dashboard
- Generate business insights

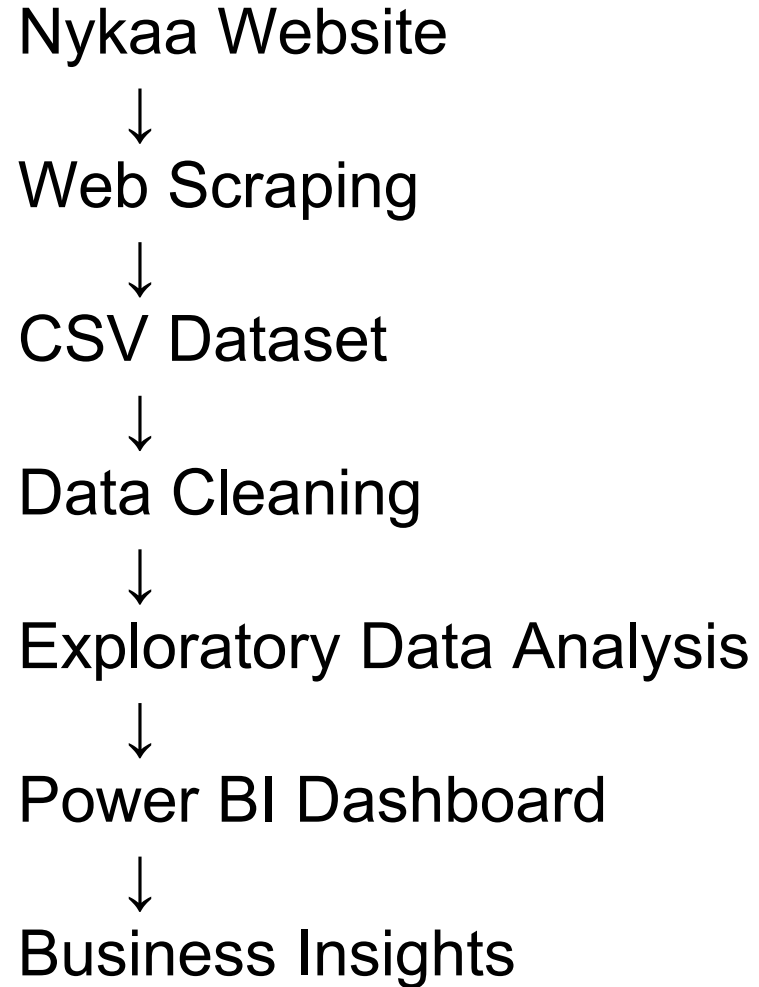
Tools & Technologies

Tools Used:

Tools	Purpose
• Python	Programming Language
• Requests	Fetch Website Data
• Beautiful Soup	Web Scraping
• Pandas	Data Cleaning & Analysis
• Num py	Numerical Computation
• Matplotlib	Data Visualization
• Seaborn	Statistical Visualization
• Jupyter Notebook	Data Analysis & EDA
• Power Bi	Dashboard Development
• CSV	Data Storage

Methodology

Project Workflow



Data Cleaning

Data Cleaning Process

- Removed duplicate records
- Handled missing values
- Corrected data types
- Standardized column names
- Removed unwanted symbols
- Converted price and rating into numeric format

Exploratory Data Analysis

Performed:

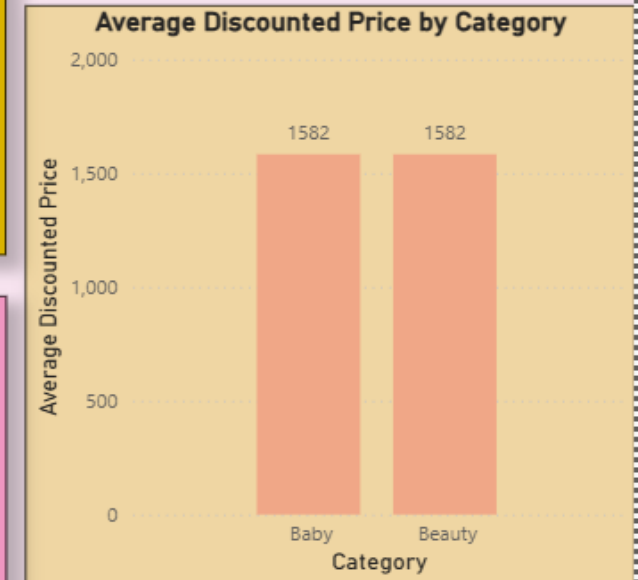
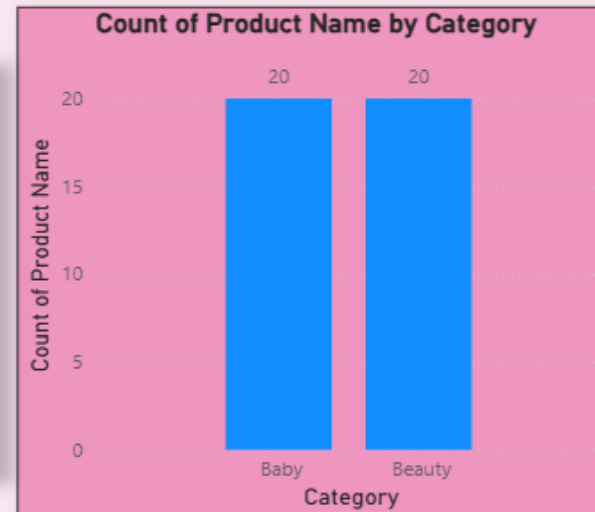
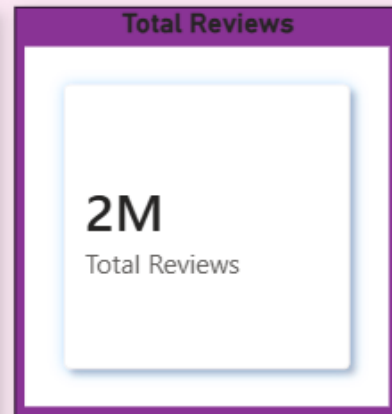
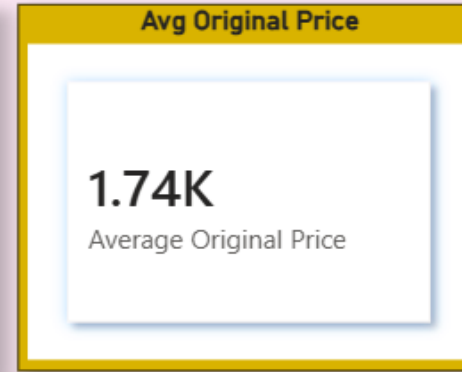
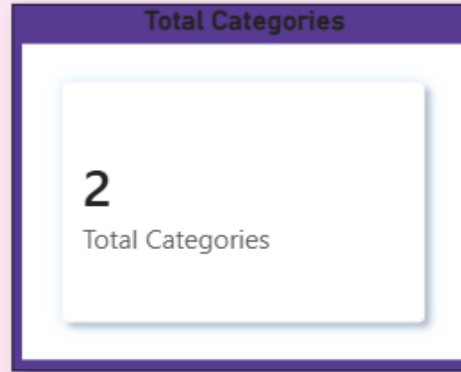
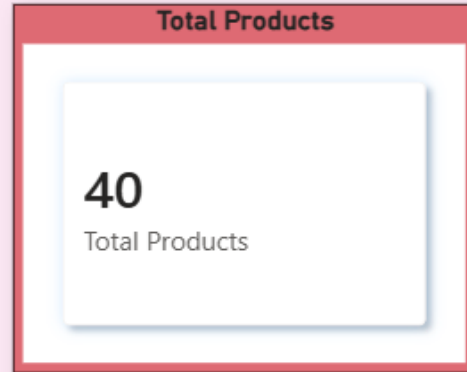
- Product Distribution
- Brand Distribution
- Price Analysis
- Rating Analysis
- Discount Analysis
- Customer Review Analysis
- Correlation Analysis

Visualizations Used :

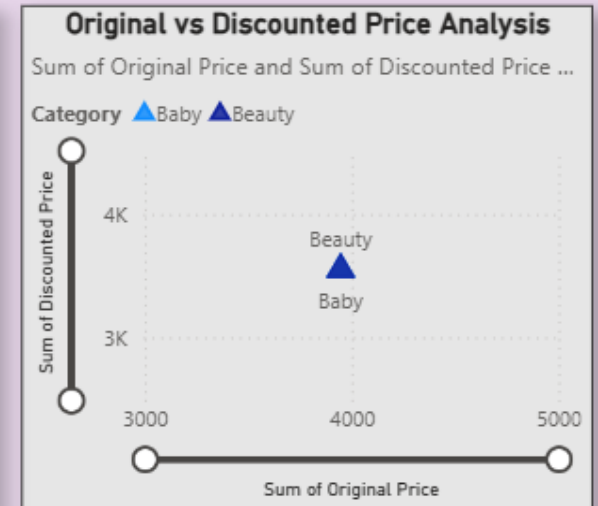
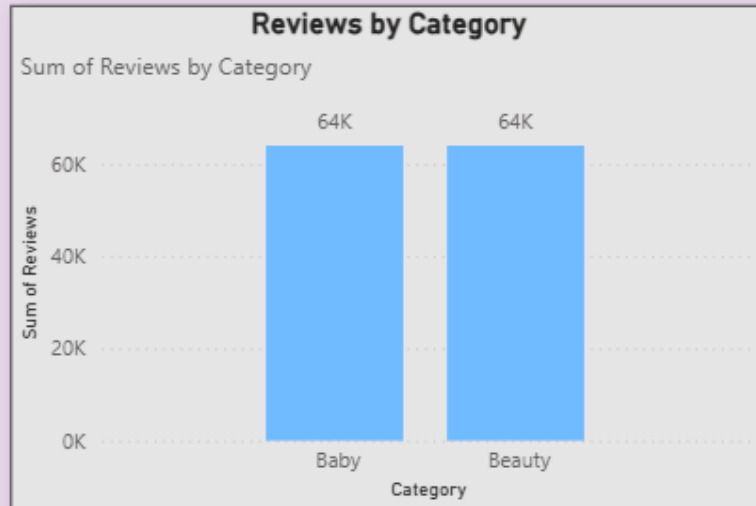
- Histogram
- Bar Chart
- Pie Chart
- Box Plot
- Scatter Plot
- Heatmap

Power BI Dashboard

Nyka Beauty & Baby Products Analysis



Product Reviews, Pricing & Discount Analysis



Category

☐ Baby

☐ Beauty

Discounted Price

115 4149

Original Price

115 5927

Discount

☒ 10% Off

☐ 15% Off

☐ 17% Off

☐ 25% Off

☐ 30% Off

☐ 5% Off

☐ No Discount

Key Insights

Key Findings :

- Beauty products form the largest portion of the dataset.
- Some brands offer significantly more products than others.
- Premium brands generally have higher prices.
- Most products have ratings above 4.
- Discount percentages vary across brands.
- Higher-rated products often receive more customer reviews.

Conclusion & Future Scope

Conclusion

- Successfully collected Nykaa product data using web scraping.
- Cleaned and transformed the dataset.
- Performed Exploratory Data Analysis.
- Developed an interactive Power BI dashboard.
- Generated meaningful business insights.

Future Scope

- Real-time web scraping
- Machine learning for price prediction
- Sentiment analysis of customer reviews
- Multi-platform product comparison
- Live Power BI dashboards

THANK
YOU

